Notebook_ca3

November 18, 2025

1 IND320 Course Project, Part 3

1.1 Code access and direct links

- The project is deployed here: ind320-henrikengdal-project
- The code is accessible at the repository: henrikengdal/ind320-henrikengdal-project

1.2 AI Usage

AI plays a multifaceted role throughout this project, primarily serving as an assistant and analytical tool. The project leverages AI in several areas:

Development and Code Generation: AI assists in writing and optimizing code for the application.

Data Analysis and Insights: AI helps analyze data patterns and identifying trends. It assists in generating meaningful statistical summaries and suggesting appropriate visualization techniques for the given data.

Documentation and Communication: AI supports the creation of clear documentation, such as code comments, and user interface text. It helps structure the project documentation and ensures technical concepts are communicated effectively.

Problem-Solving and Debugging: Throughout the development process, AI serves as a coding companion, helping troubleshoot issues, optimize data processing workflows, and suggesting best practices.

1.3 Defining representatives

[6]:
```
import pandas as pd
price_areas = {
    'NO1': ('Oslo', 10.7522, 59.9139),
    'NO2': ('Kristiansand', 7.9956, 58.1467),
    'NO3': ('Trondheim', 10.3951, 63.4305),
    'NO4': ('Tromsø', 18.9553, 69.6496),
    'NO5': ('Bergen', 5.3221, 60.3913)
}
df = pd.DataFrame.from_dict(price_areas, orient='index', columns=['City',␣
↪'Longitude', 'Latitude'])
```

1.4 API Download

[7]:
```python
import requests_cache
from datetime import datetime

# Fallback if retry_requests is not available
try:
    from retry_requests import retry
except Exception:
    def retry(session, retries: int = 0, backoff_factor: float = 0.0):
        return session

# Use ERA5 archive endpoint for historical data like 2019
BASE_URL = "https://archive-api.open-meteo.com/v1/era5"
HOUR_VARS = [
    "temperature_2m",
    "precipitation",
    "windspeed_10m",
    "windgusts_10m",
    "winddirection_10m",
]

# Cached session with retries for resilient API calls
_session = requests_cache.CachedSession(
    cache_name="openmeteo_cache", backend="sqlite", expire_after=60 * 60 * 24
)
_session = retry(_session, retries=3, backoff_factor=0.2)

def download_weather_data(longitude: float, latitude: float, year: int,␣
 ↪session=_session) -> pd.DataFrame:
    """Download hourly ERA5 weather for a given location and year.

    Parameters
    - longitude, latitude: geographic coordinates
    - year: e.g. 2019

    Returns
    - DataFrame with columns: time + requested hourly variables
    """
    start_date = f"{year}-01-01"
    end_date = f"{year}-12-31"

    params = {
        "latitude": latitude,
        "longitude": longitude,
        "start_date": start_date,
        "end_date": end_date,
```

```
        "hourly": ",".join(HOUR_VARS),
        "timezone": "Europe/Oslo",
    }

    r = session.get(BASE_URL, params=params, timeout=30)
    r.raise_for_status()
    data = r.json()

    if "hourly" not in data:
        raise ValueError(f"Unexpected API response. Top-level keys: {list(data.
↪keys())}")

    hourly = data["hourly"]
    # Build DataFrame in a robust way, filling missing variables with NA
    df = pd.DataFrame({"time": pd.to_datetime(hourly["time"])})
    for var in HOUR_VARS:
        df[var] = hourly.get(var)
    return df

# Example: Bergen 2019
df_bergen = download_weather_data(price_areas['NO5'][1], price_areas['NO5'][2],␣
↪2019)
df_bergen.head()
```

[7]:

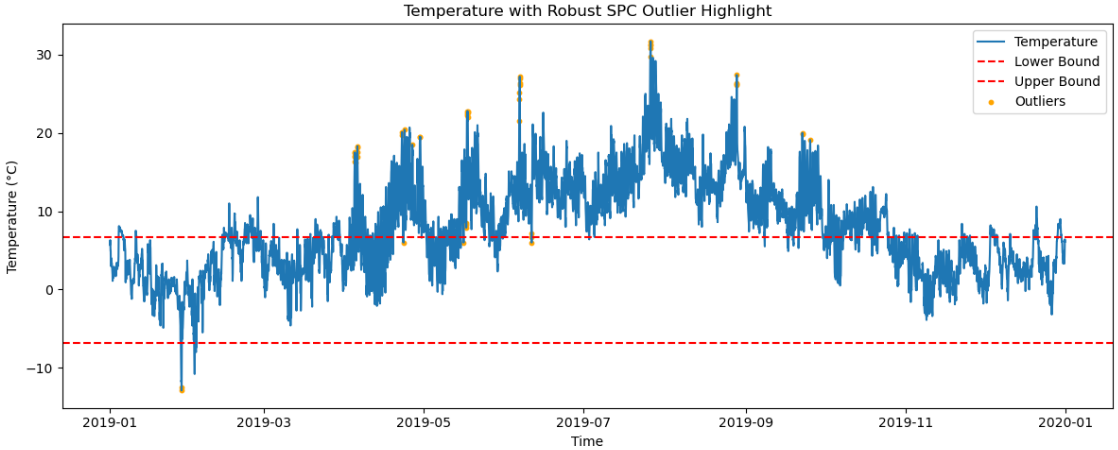
```
                 time  temperature_2m  precipitation  windspeed_10m  \
0 2019-01-01 00:00:00             5.7            0.7           37.0
1 2019-01-01 01:00:00             5.8            0.2           41.0
2 2019-01-01 02:00:00             6.1            0.7           42.0
3 2019-01-01 03:00:00             6.3            0.5           40.9
4 2019-01-01 04:00:00             5.8            1.1           41.2

   windgusts_10m  winddirection_10m
0           99.7                263
1          107.3                278
2          112.0                286
3          105.8                298
4          110.2                315
```

1.5 Outliers and anomalies

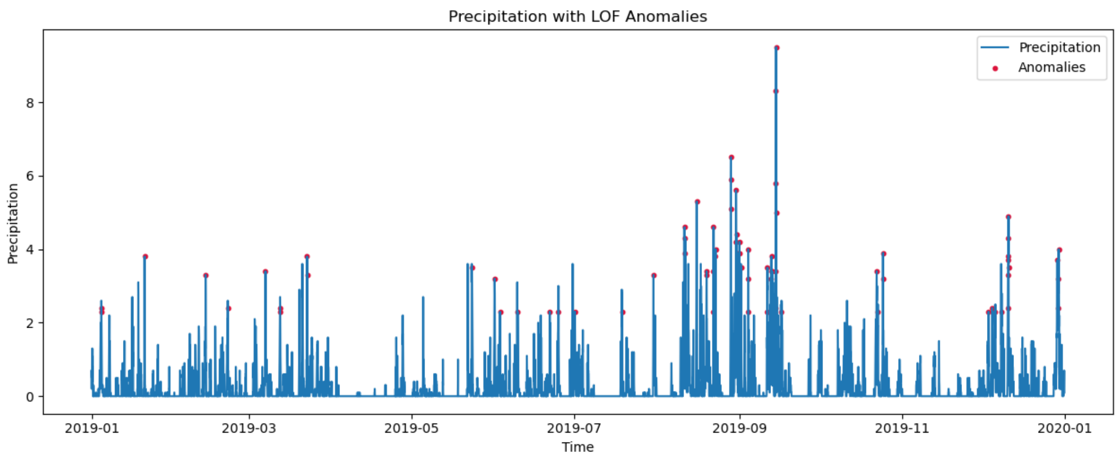
[8]:

```
import numpy as np
from sklearn.neighbors import LocalOutlierFactor
from scipy.fft import dct, idct
import matplotlib.pyplot as plt

def plot_temperature_with_outliers_matplotlib(df, dct_cutoff=200, n_std=3.5):
    temp_data = df['temperature_2m'].values
```

```python
    temp_dct = dct(temp_data, type=2, norm='ortho')
    temp_dct[:dct_cutoff] = 0
    satv = idct(temp_dct, type=2, norm='ortho')
    median_satv = np.median(satv)
    mad_scale = np.median(np.abs(satv - median_satv)) * 1.4826
    lower_bound = median_satv - n_std * mad_scale
    upper_bound = median_satv + n_std * mad_scale
    outliers = (satv < lower_bound) | (satv > upper_bound)
    outlier_indices = np.where(outliers)[0]
    plt.figure(figsize=(12, 5))
    plt.plot(df['time'], temp_data, label='Temperature', color='#1f77b4')
    plt.axhline(lower_bound, color='red', linestyle='--', label='Lower Bound')
    plt.axhline(upper_bound, color='red', linestyle='--', label='Upper Bound')
    plt.scatter(df['time'].iloc[outlier_indices], temp_data[outlier_indices],␣
 ↪color='orange', label='Outliers', s=10)
    plt.title('Temperature with Robust SPC Outlier Highlight')
    plt.xlabel('Time')
    plt.ylabel('Temperature (°C)')
    plt.legend()
    plt.tight_layout()
    plt.show()

def plot_precipitation_with_anomalies_matplotlib(df, contamination=0.01,␣
 ↪n_neighbors=20):
    precip_data = df['precipitation'].values.reshape(-1, 1)
    lof = LocalOutlierFactor(n_neighbors=n_neighbors,␣
 ↪contamination=contamination)
    anomaly_labels = lof.fit_predict(precip_data)
    anomaly_mask = anomaly_labels == -1
    anomaly_indices = np.where(anomaly_mask)[0]
    plt.figure(figsize=(12, 5))
    plt.plot(df['time'], df['precipitation'], label='Precipitation',␣
 ↪color='#1f77b4')
    if len(anomaly_indices) > 0:
        plt.scatter(df['time'].iloc[anomaly_indices], df['precipitation'].
 ↪iloc[anomaly_indices], color='crimson', label='Anomalies', s=10)
    plt.title('Precipitation with LOF Anomalies')
    plt.xlabel('Time')
    plt.ylabel('Precipitation')
    plt.legend()
    plt.tight_layout()
    plt.show()

# Test the functions with Bergen data
plot_temperature_with_outliers_matplotlib(df_bergen)
plot_precipitation_with_anomalies_matplotlib(df_bergen)
```




```
/Users/henrikengdal/Documents/GitHub/IND320-henrikengdal-
project/.conda/lib/python3.11/site-packages/sklearn/neighbors/_lof.py:322:
UserWarning:

Duplicate values are leading to incorrect results. Increase the number of
neighbors for more accurate results.
```

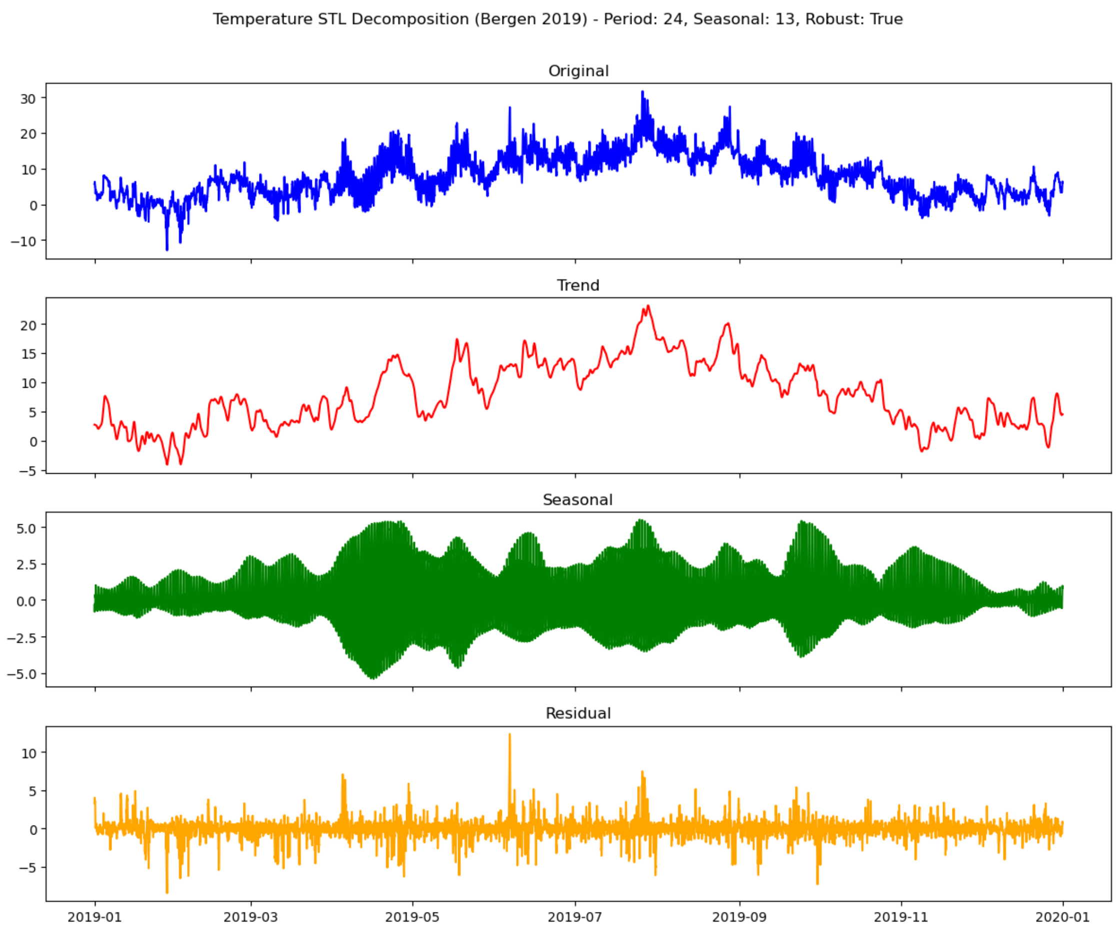


1.6 Seasonal-Trend decomposition using LOESS (STL)

[11]:
```
from statsmodels.tsa.seasonal import STL
import matplotlib.pyplot as plt
import pandas as pd
```

```
def plot_stl_decomposition_matplotlib(df, value_col='production_mwh',␣
↪time_col='time',
                          period=24*7, seasonal=7, trend=None, robust=True,
                          title_prefix="STL Decomposition"):
    """
    Perform STL decomposition on time series data and create a Matplotlib plot.
    """
    # Set time as index if not already
    if time_col in df.columns:
        ts_data = df.set_index(time_col)[value_col]
    else:
        ts_data = df[value_col]
    ts_data = ts_data.dropna()
    period_int = int(round(period))
    if period_int < 2:
        raise ValueError(f"period must be >= 2, got {period}")
    def _ensure_odd(n):
        if n is None:
            return None
        n_int = int(round(n))
        return n_int if n_int % 2 == 1 else n_int + 1
    seasonal_odd = _ensure_odd(seasonal)
    trend_odd = _ensure_odd(trend)
    stl = STL(ts_data, period=period_int, seasonal=seasonal_odd,␣
↪trend=trend_odd, robust=robust)
    stl_result = stl.fit()
    # Plot with matplotlib
    fig, axs = plt.subplots(4, 1, figsize=(12, 10), sharex=True)
    axs[0].plot(stl_result.observed.index, stl_result.observed.values,␣
↪color='blue')
    axs[0].set_title('Original')
    axs[1].plot(stl_result.trend.index, stl_result.trend.values, color='red')
    axs[1].set_title('Trend')
    axs[2].plot(stl_result.seasonal.index, stl_result.seasonal.values,␣
↪color='green')
    axs[2].set_title('Seasonal')
    axs[3].plot(stl_result.resid.index, stl_result.resid.values, color='orange')
    axs[3].set_title('Residual')
    plt.suptitle(f"{title_prefix} - Period: {period_int}, Seasonal:␣
↪{seasonal_odd}, Robust: {robust}")
    plt.tight_layout(rect=[0, 0, 1, 0.97])
    plt.show()
    return stl_result

# Test the function with Bergen temperature data (use daily seasonality)
stl_temp_result = plot_stl_decomposition_matplotlib(
    df_bergen,
```

```
    value_col='temperature_2m',
    time_col='time',
    period=24,          # Daily cycle for hourly data
    seasonal=13,        # Seasonal smoother length (odd)
    title_prefix="Temperature STL Decomposition (Bergen 2019)"
    )

# Display decomposition statistics
print("STL Decomposition Results:")
print(f"Seasonal strength: {1 - stl_temp_result.resid.var() / (stl_temp_result.
↪seasonal + stl_temp_result.resid).var():.3f}")
print(f"Trend strength: {1 - stl_temp_result.resid.var() / (stl_temp_result.
↪trend + stl_temp_result.resid).var():.3f}")
```



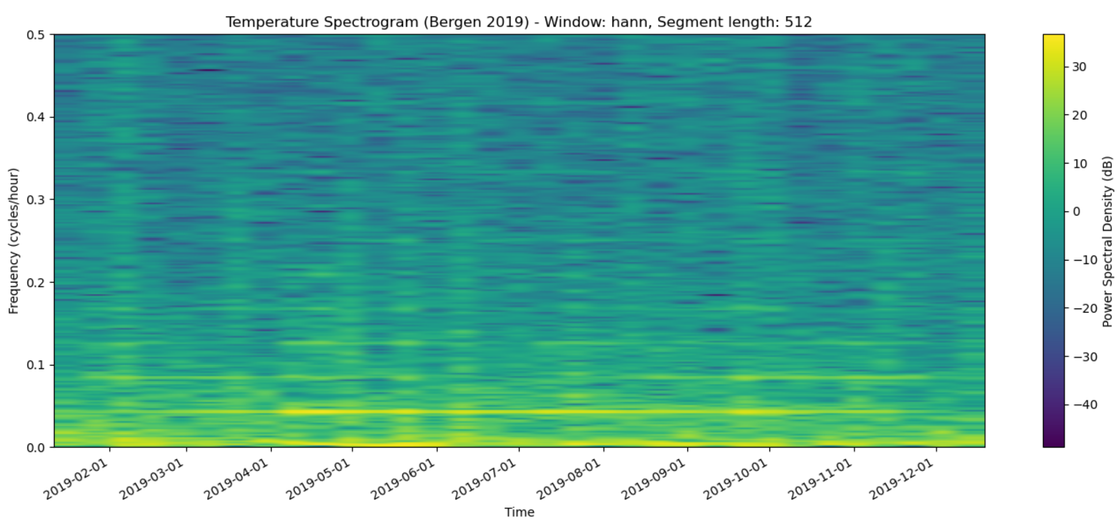

```
STL Decomposition Results:
Seasonal strength: 0.670
Trend strength: 0.951
```

1.7 Spectrogram

[13]:
```python
import numpy as np
from scipy import signal
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.dates as mdates

def create_spectrogram_matplotlib(df, value_col='production_mwh',␣
 ↪time_col='time',
                              window='hann', nperseg=256, noverlap=None,
                              title_prefix="Spectrogram"):
    """
    Create a spectrogram visualization for time series data using Matplotlib.

    Parameters:
    - df: DataFrame with time series data
    - value_col: Column name containing the values to analyze
    - time_col: Column name containing datetime values
    - window: Window function ('hann', 'hamming', 'blackman', etc.)
    - nperseg: Length of each segment (window size)
    - noverlap: Number of points to overlap between segments (default: nperseg//
 ↪8)
    - title_prefix: Prefix for plot title

    Returns:
    - spec_data: Dictionary with spectrogram data (frequencies, times, Sxx)
    """
    # Set default overlap if not provided
    if noverlap is None:
        noverlap = nperseg // 8

    # Extract time series data
    if time_col in df.columns:
        ts_data = df.set_index(time_col)[value_col]
    else:
        ts_data = df[value_col]

    # Remove NaN values
    ts_data = ts_data.dropna()

    # Calculate sampling frequency (assuming regular intervals)
    time_diff = ts_data.index.to_series().diff().median()
    fs = 1 / (time_diff.total_seconds() / 3600)  # Convert to samples per hour

    # Compute spectrogram
    frequencies, times, Sxx = signal.spectrogram(
```

```
        ts_data.values,
        fs=fs,
        window=window,
        nperseg=nperseg,
        noverlap=noverlap,
        scaling='density'
    )

    # Convert power spectral density to dB
    Sxx_db = 10 * np.log10(Sxx + 1e-10)  # Add small value to avoid log(0)

    # Convert time segments back to datetime
    start_time = ts_data.index[0]
    time_segments = [start_time + pd.Timedelta(hours=t) for t in times]

    # Create spectrogram plot with matplotlib
    fig, ax = plt.subplots(figsize=(14, 6))
    extent = [mdates.date2num(time_segments[0]), mdates.date2num(time_segments[-1]), frequencies[0], frequencies[-1]]
    im = ax.imshow(Sxx_db, aspect='auto', origin='lower', extent=extent, cmap='viridis')
    cbar = fig.colorbar(im, ax=ax, label="Power Spectral Density (dB)")
    ax.set_title(f"{title_prefix} - Window: {window}, Segment length: {nperseg}")
    ax.set_xlabel("Time")
    ax.set_ylabel("Frequency (cycles/hour)")
    ax.xaxis_date()
    ax.xaxis.set_major_formatter(mdates.DateFormatter('%Y-%m-%d'))
    fig.autofmt_xdate()
    plt.tight_layout()
    plt.show()

    # Prepare return data
    spec_data = {
        'frequencies': frequencies,
        'times': times,
        'time_segments': time_segments,
        'Sxx': Sxx,
        'Sxx_db': Sxx_db,
        'sampling_freq': fs,
        'window': window,
        'nperseg': nperseg,
        'noverlap': noverlap
    }

    return spec_data
```

```python
# Test the function with Bergen temperature data
spec_data = create_spectrogram_matplotlib(
    df_bergen,
    value_col='temperature_2m',
    time_col='time',
    window='hann',
    nperseg=512,  # Larger window for better frequency resolution
    noverlap=256,  # 50% overlap
    title_prefix="Temperature Spectrogram (Bergen 2019)"
)

# Display spectrogram statistics
print("Spectrogram Analysis Results:")
print(f"Sampling frequency: {spec_data['sampling_freq']:.2f} samples/hour")
print(f"Frequency resolution: {spec_data['frequencies'][1] - spec_data['frequencies'][0]:.4f} cycles/hour")
print(f"Time resolution: {spec_data['times'][1] - spec_data['times'][0]:.2f} hours")
print(f"Number of frequency bins: {len(spec_data['frequencies'])}")
print(f"Number of time segments: {len(spec_data['times'])}")
print(f"Power range: {spec_data['Sxx_db'].min():.1f} to {spec_data['Sxx_db'].max():.1f} dB")
```




```
Spectrogram Analysis Results:
Sampling frequency: 1.00 samples/hour
Frequency resolution: 0.0020 cycles/hour
Time resolution: 256.00 hours
Number of frequency bins: 257
Number of time segments: 33
Power range: -48.8 to 36.6 dB
```

1.8 Word log

For this part, I found myself using AI more frequently to understand what was being asked for in the CA description than I did with the previous two parts. I think this was mainly because the expectations were phrased in a way that required some interpretation, and AI tools like ChatGPT helped me clarify the intent before I started implementing anything. Either way, I think we've arrived at something that covers all the necessary points and demonstrates a clear understanding of the task requirements.

Starting with the Streamlit experience, this section was relatively straightforward, but it did include a few challenges that required trial and error. The most difficult part was understanding how to reorder and manage the pages correctly while maintaining consistent naming and navigation. Once I figured that out, the implementation process went much more smoothly. The "new A" page was implemented without major issues. I used GitHub Copilot to get a sense of what the "necessary UI elements" might look like, and it provided good suggestions that aligned closely with what I needed. There were still a few moments where I had to manually adjust the layout or functionality, but overall the experience was productive.

The "new B" page, however, brought a few challenges, especially with implementing caching for selections made in the "Energy Production Analysis" page. It took some testing and revisions to get everything working as intended, but I'm satisfied with the final behavior.

The notebook part of the assignment was also quite manageable. Since this section required creating and refining several methods, I used AI support mainly for generating documentation and improving code readability. This not only saved time but also helped maintain consistency across the different components of the project.

Overall, I gained a better understanding of how to structure the Streamlit application effectively.